MIL-C-14522D
20 March 1987
SUPERSEDING
MIL-C-14522C(ME)
12 January 1979

MILITARY SPECIFICATION

COMPONENTS, PREFABRICATED STRONG ROOM, PORTABLE: FOR SMALL ARMS AND EQUIPMENT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers portable, prefabricated strong room components, for small arms and equipment. It includes all structural components, clamps, and fasteners necessary to assemble strong rooms of various sizes as illustrated in figure 1.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

2.1.1 Documents. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5450

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SPECIFICATIONS

FEDERAL

FF-S-325	- Shield, Expansion, Nail, Expansion, and Nail, Drive Screw (Devices, Anchoring, Masonry)
QQ-S-781	- Strapping, Steel, and Seals
PPP-B-601	- Boxes, Wood, Cleated-Plywood

MILITARY

MIL-T-704	- Treatment and Painting of Materiel
MIL-C-3774	- Crate, Wood, Open, 12,000 and 16,000 Pound Capacity
MIL-C-52950	- Crates, Wood, Open and Covered

STANDARDS

FEDERAL

FED-STD-595	- Colors

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MS16992	- Bolt (Screw), Lag, Hex Head, Gimlet Point

* 2.1.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWING

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

13218E8075	- Strong Room, Prefabricated Portable

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

* 2.2 Other publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in

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the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

AMERICAN WELDING SOCIETY, INC. (AWS)

A5.1 - Specification for Covered Carbon Arc-Welding Electrodes
A5.18 - Specification for Carbon Steel Filler Metals for Gas Shielded Arc Welding
D1.1 - Structural Welding Code for Steel

(Application for copies should be addressed to the American Welding Society, P.O. Box 351040, 550 LeJeune Road, Miami, FL 33135.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding and Brazing Qualifications

(Application for copies should be addressed to the American Society of Mechanical Engineers, 345 East 47th Street, New York, NY 10017.)

STEEL STRUCTURES PAINTING COUNCIL (SSPC)

SSPC-SP 1 - Commercial Blast Cleaning

(Application for copies should be addressed to the Steel Structures Painting Council, 4400 Fifth Avenue, Pittsburgh, PA 15213.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Technical society and technical association documents are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document shall take precedence. Nothing in this document, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample of each component shall be subjected to first article inspection (see 4.3, 6.2, and 6.3).

3.2 Materials and components. The materials and components shall be as specified herein and as shown on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturer provided that the completed item complies with all the provisions of this document (see 6.4).

3.2.1 Welding electrodes. Welding electrodes shall conform to AWS A5.1 or AWS A5.18. Unless otherwise specified, the class of electrodes used will be at the contractor's option.

3.2.2 Wire mesh. Wire mesh shall be as specified in figure 2.

3.2.3 Wall and floor clamp. When the wall and floor clamp is specified (see 6.2), it shall be furnished with a lag bolt conforming to MS16992-543. When the clamp is to be applied to concrete or masonry surfaces, an expansion anchor shield conforming to FF-S-325, Type 1, Class 1, Group II, 5/16 inch diameter shall be furnished with each clamp.

3.3 Design. Components for prefabricated strong rooms shall be in accordance with Drawing 13218E8075 (see 6.2) and as specified herein. The drawings forming a part of this document is an end product drawing. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to ensure correct fit, assembly, and operation of the strong room components. Any data (e.g., shop drawings, layouts, flow sheets, processing procedures) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or his designated representative.

3.4 Identification marking. All strong room components shall be metal stamped as shown on the drawings. All metal stamping shall be legible after painting.

3.5 Finish.

3.5.1 Cleaning. After fabrication, each strong room component shall be prepared for painting in accordance with SSPC-SP 1.

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3.5.2 Treatment and painting. Each component shall be treated and painted in accordance with MIL-T-704 except that, unless otherwise specified (see 6.2), the finish coat shall be semigloss black enamel, Color No. 27038 in accordance with FED-STD-595.

3.6 Workmanship.

3.6.1 Casting. All cast strong room components shall be free from harmful extraneous material such as sand, dirt, pits, sprues, scale and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

3.6.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material; corners shall be square and true. Flame cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing, providing all slag is removed after burning. All bends shall be made with controlled means to ensure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.6.3 Machine work. Tolerances and gages for metal fits shall conform to the limits specified on the drawings and to the standard of the industry.

3.6.4 Welding. The surfaces to be welded shall be free from rust, scale, paint, grease, and other foreign matter. Welds shall transmit stress without permanent deformation or failure and shall have no crack type defects when the component is completed. Parent materials, weld filler metals, and fabrication techniques shall be as required to enable the components to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 1/16 inch, unless appropriate bridging techniques are used. Unless otherwise specified, the welding process used in fabrication of the component shall be at the option of the contractor.

3.6.5 Welders. Before assigning any welder to welding work covered by this document, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

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AWS D1.1, Structural Welding Code for Steel

Welding and Brazing Qualification of the ASME Boiler and Pressure Vessel Code (Section IX).

Contractors who only make horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds. For the purpose of this document, automatic welding equipment operators shall be considered welders and subject to the qualification procedures outlined above.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirement in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

* 4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

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4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3. The presence of any defect shall be cause for rejection of the first article.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

* 4.4.1.1 Certification. The contractor shall furnish a certificate of compliance for the requirements specified in 3.2.1 and 3.6.5.

* 4.4.2 End item visual examination. Each type of component shall be examined for the applicable defects listed below. The lot size shall be expressed in units of all strong room components. The sample unit shall be one component. The percentage of samples of each component in relation to the total of all components sampled shall be proportional to the percentage of that type of component in the lot in relation to the total of all components in the lot. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Material	Not as specified
Wire mesh	Not as specified
Wall and floor clamp	Not as specified, when specified Anchor shield missing or not as specified
Components	Missing or not as specified
Identification marking	Not as specified

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Examine	Defect
Finish	Cleaning, treatment, or painting not as specified
Workmanship	Not as specified

* 4.4.3 End item dimensional examination. Each type of component shall be examined for conformance to the dimensions annotated with an asterisk on the drawings. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of all strong room components. The sample unit shall be one component. The percentage of samples of each component in relation to the total of all components sampled shall be proportional to the percentage of that type of component in the lot in relation to the total of all components in the lot. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Components of unlike description packed or bundled together, except for consolidated packing Consolidated packing not as specified

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5. PACKAGING

5.1 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.1.1 Level A packing.

5.1.1.1 Spacers and clamps. Spacers and clamps of like description shall be packed in boxes conforming to overseas type, style optional of PPP-B-601. Boxes exceeding 250 pounds shall be modified with skids in accordance with PPP-B-601. Strapping shall conform to Class 1, Type I or IV, size as applicable, finish A of QQ-S-781. When skids are furnished, strapping shall be placed adjacent to the skids. When consolidated packing is specified (see 5.1.1.4), the boxes shall not require skids or strapping.

5.1.1.2 Angles and channels. Angles and channels of like description shall be bundled in quantities most convenient for packing. The bundles shall be secured with not less than two flat steel straps conforming to class 1, type I or IV, finish B of QQ-S-781. The bundles shall be packed together in open wood crates conforming to style A, type II of MIL-C-52950. Blocking and bracing shall be in accordance with the appendix to MIL-C-52950. When consolidated packing is specified (see 5.1.1.4), the crating shall not be required.

5.1.1.3 Floor plates and panels. Floor plates and panels shall be packed together in open crates conforming to style A, type II of MIL-C-52950. Blocking and bracing shall be in accordance with the appendix to MIL-C-52950. When consolidated packing is specified (see 5.1.1.4), the crating shall not be required.

5.1.1.4 Consolidated packing. When specified (see 6.2), spacers and clamps packed as specified in 5.1.1.1, the angles and channels bundled as specified in 5.1.1.2, and the uncrated floor plates and panels shall be packed together in a crate conforming to skid-type base, nailed assembly of MIL-C-3774, or in a crate conforming to style A, type V, nondemountable of MIL-C-52950. Blocking, bracing, and anchoring shall be provided to prevent shifting of and damage to the contents and shall be in accordance with the appendix to the applicable crate document.

5.1.2 Level B packing. The components shall be packed as specified for level A, except boxes for spacers and clamps shall be domestic type.

5.1.3 Commercial packing. Components shall be packed in accordance with ASTM D 3951.

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5.2 Marking. Unless otherwise specified in the contract or purchase order, marking shall be in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

6.1 Intended use. Each component is intended for use in the construction of various sizes of strong rooms. These strong rooms are used for the safe storage of small arms and equipment in National Guard Armories and similar installations.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.1, 4.3, and 6.3).
c. When the wall and floor clamp is required and when the anchor shield is required (see 3.2.3).
d. Components required and quantity of each component required (see 3.3).
e. Color required when other than as specified (see 3.5.2).
f. Level of packing required (see 5.1).
g. When consolidated packing is required (see 5.1.1.4).

* 6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this document (see 3.2).

* 6.5 Subject term (key word) listing.

Arms, small
Storage, equipment
Strong room components

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6.6 Changes from previous issue. The margins of this document are marked with an asterisk (*) to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only, and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, as written, irrespective of the marginal notations and relationship to the last previous issue.

Custodians:

Army - GL
Air Force - 99

Review activities:

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Preparing activity:

Army - GL

Project No. 5450-0022

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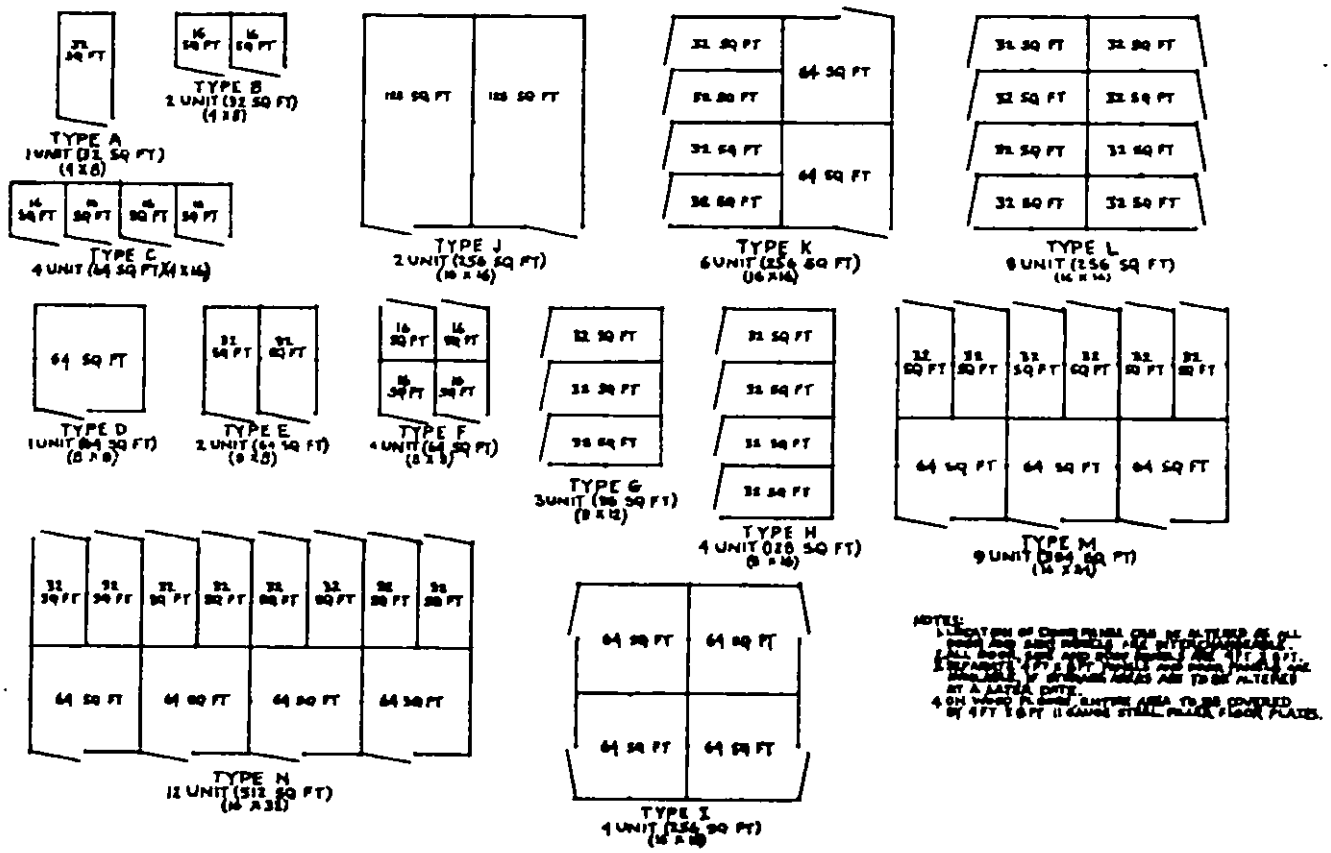
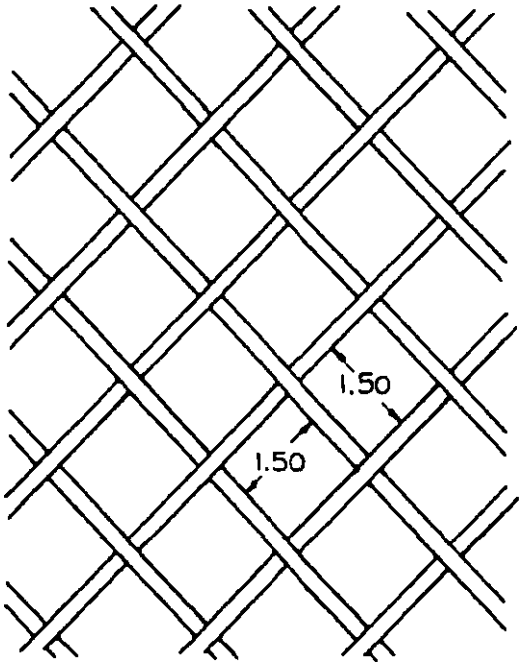
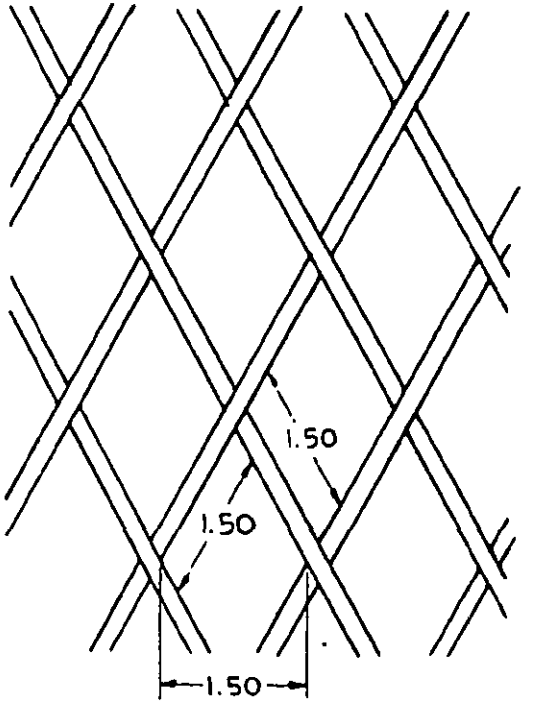


FIGURE 1. Typical strong room combinations.

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SQUARE DIAMOND



LONG DIAMOND

NOTES:
1. WIRE TO BE .160-.165 DIA, 1035 TO 1040 STEEL.
2. MANUFACTURERS STANDARD COMMERCIAL TOLERANCES SHALL APPLY.

X - 2111A

FIGURE 2. Diamond wire mesh.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-14522D	Components, Prefabricated Strong Room, Portable: For Small Arms And Equipment

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

DD FORM 82 MAR 1426 PREVIOUS EDITION IS OBSOLETE.

NATICK OP, 1 Apr 84

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